APPENDIX

A. Baseline Details

The baselines adopted in this study fall into three categories: (1) contrastive-based approaches (Moco-v3, CLOCS, Graph-CL, CLEF); (2) generative-based approaches (CRT, ST-MEM, TimesURL); and (3) multimodal fusion-based approach (MERL, ESI).

- Moco-v3 [5] leverages momentum-updated dual encoders and vision transformer backbones to learn high-quality representations without labels. By eliminating the memory queue used in earlier versions, it simplifies contrastive learning while maintaining strong performance across visual and time-series tasks.
- CLOCS [6] considers different ECG segments or lead-wise views from the same patient as positive samples, utilizing a convolutional neural network backbone to learn discriminative and patient-specific ECG representations.
- CLEF [7] is a contrastive learning method that uses dynamically weighted negative samples from clinical risk scores to guide the model in learning meaningful ECG representations through clinical metadata, thereby improving the performance of downstream tasks without the need for manual annotation.
- Graph-CL [8] models ECG as graphs, and it creates augmented views of input graphs via structure-preserving transformations (e.g., edge perturbation, node dropping), and maximizes agreement between these views using a contrastive loss.
- CRT [9] leverages a Transformer architecture to reconstruct masked patches across time and frequency domains, enabling joint temporal-spectral representation learning through cross-domain reconstruction.
- ST-MEM [10] incorporates a lead-wise shared decoder to model spatial relationships, and employs a vision transformer as the encoder to reconstruct masked 12-lead ECG signals for spatiotemporal feature learning.
- TimesURL [11] introduce a frequency-temporal-based augmentation to keep the temporal property unchanged, and time reconstruction as a joint optimization objective with contrastive learning is used to capture both segment-level and instance-level information.
- MERL [12] achieves zero-shot ECG classification by learning multimodal representations of paired ECG and reports via a simple dropout strategy, and enhances textual prompts with clinical knowledge via LLM.
- ESI [13] employs a LLM based on RAG to generate detailed textual descriptions of ECGs and uses a dual-objective loss function to constrain the ECG representations.

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